Magnitude, Not Shape: Volatility Beats Downside- and Tail-Risk Metrics at Forecasting Drawdown

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ABSTRACT

Every few years a new downside- or tail-risk metric promises to improve on plain return volatility. The premise is always the same. The *shape* of the loss distribution, whether its asymmetry, tail heaviness, or co-crash structure, is supposed to carry information that symmetric volatility misses. We test that premise as adversarially as the data allow. On a survivorship-free, multi-regime, multiple-testing-corrected evaluation of ten metrics across five test beds, no metric reliably beats volatility at forecasting an asset's forward maximum drawdown, whether the metric is published or machine-discovered. The failure has a clean anatomy, which we call *magnitude, not shape*. The partial rank correlation of Value-at-Risk with forward drawdown, controlling for volatility, survives Benjamini–Hochberg correction on all five beds, though it reaches only 0.19 at its largest. Every tail-shape metric adds nothing or subtracts. Yet the shape metrics are not broken. On genuinely shape-sensitive targets they beat volatility. Drawdown is simply a magnitude phenomenon that shape does not drive. Volatility is hard to beat because it is persistent, but a predictive mediation test indicates it is not a sufficient statistic. The tiny magnitude edge is not economically free either. A cost-aware backtest and a pre-registered composite add nothing over volatility except in fat-tailed crypto. Finally, an autonomous large-language-model discovery loop only inflates validation while its locked-test score plateaus at volatility, a data-snooping caution now that agentic search makes signal-hunting nearly free. Volatility is not glamorous, but on a level field it is very hard to beat.

CCS CONCEPTS

• **Computing methodologies → Machine learning**; • **Applied computing → Economics**.

KEYWORDS

downside risk, tail risk, drawdown, volatility, survivorship bias, multiple testing, data snooping, LLM agents

1 INTRODUCTION

Prices fall faster than they rise, and it is the fall that ruins investors. From that intuition a large literature has grown. It proposes metrics that look past symmetric volatility to the shape of the loss distribution. These include lower partial moments and the Sortino ratio [6, 15, 22], the partial-moment "predictive" ratio (UPM/LPM) [23], left-tail Value-at-Risk and expected shortfall [3, 21], realized semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment-aversion risk (GDA) [13], and tail-exponent factors [17]. Each rests on the same premise, that asymmetry or tail structure carries information volatility misses.

We test that premise on the fairest field we can build. Three disciplines the source literature usually omits turn out to matter enormously. A *survivorship-free* universe keeps the assets that actually died. A *locked* out-of-sample holdout stops search from leaking into the verdict. And *multiple-testing correction* keeps an evaluation of dozens of comparisons from manufacturing winners. We evaluate ten metrics, volatility and nine downside- and tail-risk challengers, as forecasters of 90-day forward maximum drawdown across five cross-sectional test beds spanning calm, bull, and crash regimes in crypto and equity, with per-date rank correlations, paired block-bootstrap confidence intervals, and Benjamini–Hochberg (BH) correction.

The headline is negative and, in hindsight, unsurprising. No metric reliably beats plain volatility. Our contribution is the anatomy of that result and the integrity of the evaluation behind it. First, a clean empirical law we call magnitude, not shape. The only signal surviving correction beyond volatility is the *size* of tail losses (VaR/ES/downside deviation), not their asymmetry (§4). Second, volatility is hard to beat because it is persistent, though a predictive test indicates forward volatility is not a sufficient statistic (§5). Third, the residual edge is not economically free, except in fat-tailed crypto (§5). Fourth, an autonomous large-language-model (LLM) discovery loop does not beat volatility out-of-sample and only overfits validation, a data-snooping caution as agentic search makes signal-hunting nearly free (§5).

Related work. Lower partial moments generalize semivariance [6, 15] and underpin the Sortino ratio [22] and the partial-moment ratio [23]; tail measures include VaR/ES [3, 21], semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment aversion [13], and tail exponents [17]. Most were built to price the cross-section, not forecast one asset's drawdown; a separate literature studies the drawdown statistic directly [10, 19]. Volatility itself is a persistent, tradable risk signal [14, 20]. That volatility is persistent is the founding fact of ARCH/GARCH [1, 7, 11, 12]; that searching many signals inflates in-sample rank correlations is classical data snooping [4, 5, 16, 24]. Our novelty is to show an *autonomous* agent reaches the snooping frontier at near-zero cost.

2 EXPERIMENTAL SETUP

Five test beds. *Crypto (survivorship-free):* a CoinMarketCap daily panel (2013–2018) that retains crashed and delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes: 2016 (calm), 2017 (bull), 2018 (crash). *Equity:* S&P 500 constituents, 1994–1999 (calm) and 2005–2009 (GFC). The equity beds are survivorship-biased by construction (freely available equity data retains almost no true deaths, itself a telling finding), so the survivorship claim rests on the crypto beds, while equity tests cross-asset generality. We checked this directly. A public, delisting-reputed price dump (the Kaggle Huge Stock Market dataset) contains none of our windows' famous S&P 500 casualties, with Lehman, Bear Stearns, Enron, and WorldCom all absent, and yields zero mid-window delistings, so no free-data equity bed is genuinely survivorship-free.[1]

What we forecast. From a 180-day trailing window we forecast each asset's 90-day forward maximum drawdown, coding a delisting as a −100% loss. We score a metric by its per-date cross-sectional rank correlation with that drawdown, averaged over the bed, cross-sectional rather than time-series because the operational question is which assets to derisk on a given date.

Metrics. All ten are computed from the same 180-day trailing daily returns. Volatility is the return standard deviation. Three are tail-magnitude measures: historical 5% Value-at-Risk and expected shortfall [21], and downside deviation, the lower partial moment of order two about a zero target [6, 22]. The rest are tail-shape measures: the Viole–Nawrocki upper/lower partial-moment ratio [23]; a per-asset Hill loss-tail exponent, our per-date simplification of the Kelly–Jiang common tail factor [17]; and four systematic co-movement measures, namely negative realized semibeta [8], downside beta [2], lower-tail dependence [9], and a disappointment-aversion proxy [13], each estimated against the equal-weight market. The partial rank correlation $\rho(\cdot, \text{drawdown} \mid \text{volatility})$ is the Spearman correlation of the two residual ranks after regressing the metric and the drawdown on volatility, per date.

Significance and power. We attach a paired block-bootstrap 95% confidence interval (length-3 blocks over the per-date series) to every comparison and apply BH correction at $q = 0.05$ within each metric family (head-to-head and partials, ~45 cells each). The crypto beds are thin (7–9 formation dates), so we read crypto-only significance as suggestive and back each crypto claim with its minimum detectable effect (MDE): VaR's increment exceeds the 80%-power MDE on crypto-16 and crypto-17 but falls below it on crypto-18. The equity beds are well powered. There the 80%-power MDE for the shape increments is 0.034–0.042, and the observed shape increments sit at or below it, so "shape adds nothing" is a powered null rather than a failure to detect. A placebo that permutes forward drawdown within each date drives the volatility correlation to ≈ 0 on four of five beds.

3 THE VERDICT: NOTHING BEATS VOLATILITY

The verdict is blunt. No metric beats volatility in a way that survives a change of asset class. Table 1 ranks each metric against volatility standalone. The only metrics that significantly beat volatility on any bed are two volatility variants, downside deviation and VaR, on the single crypto-bull bed, and both reverse on equity, where volatility significantly beats them. Every genuinely different, shape-based metric is significantly worse than volatility on every bed.

We fix the bar in advance to avoid a moving goalpost. An edge counts as *generalizable* only if a metric BH-significantly beats volatility on at least one crypto and one equity bed, i.e., survives a change of asset class. No metric clears it. Small, regime-specific edges exist for volatility's own variants, but none generalizes. We pre-registered the main tests, fixing them before we saw any results so we cannot cherry-pick them, namely the verdict, the magnitude-versus-shape increments, and the composite. The finer slices by market stress and across beds were explored afterward and count only as suggestive.

[1] All code, data pointers, and result logs to reproduce every table and figure are available in an anonymized repository for review and will be released publicly upon acceptance. Data sources are a CoinMarketCap daily crypto dump (retaining delisted coins), yfinance S&P 500 constituents and the market index, and point-in-time index membership.

Table 1: Standalone out-of-sample drawdown-forecast Spearman correlation. ∗ = significantly beats volatility, † = significantly worse (both survive BH, $q = 0.05$); unmarked = indistinguishable. The Hill tail exponent [17] is omitted here (not estimable per date on equity); its increment appears in Table 2. Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
volatility (base)	**0.62**	0.36	0.38	**0.61**	**0.56**
downside deviation	**0.62**	0.38∗	**0.41**	0.58†	0.54†
VaR	0.61	**0.40**∗	0.36	0.58†	0.54†
ES [21]	**0.62**	0.35	0.33†	0.58†	0.52†
neg. semibeta [8]	0.40†	0.26†	0.31†	0.49†	0.46†
downside beta [2]	0.01†	0.09†	0.15†	0.24†	0.34†
lower-tail dep. [9]	−0.38†	−0.15†	−0.29†	−0.07†	−0.05†
GDA [13]	−0.06†	−0.11†	−0.05†	−0.24†	−0.32†
UPM/LPM [23]	−0.55†	−0.28†	−0.35†	−0.43†	−0.42†

4 THE ANATOMY: MAGNITUDE, NOT SHAPE

If nothing beats volatility, what is left to forecast beyond it? The *partial* rank correlation (Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$, which isolates what a metric adds beyond volatility) answers this. Figure 1 plots it for every metric and bed, and the picture splits cleanly by *what a metric measures.* Tail-magnitude measures carry a small but genuine increment (VaR significant on **all five beds**, partial ρ up to 0.19; downside deviation on the three crypto beds), while tail-shape measures (the Viole–Nawrocki ratio, lower-tail dependence, the tail exponent, disappointment aversion) are ≈ 0 or negative everywhere. This is not merely an *own-asset vs systematic* artifact. The one own-asset shape metric (the Hill tail exponent) is itself ≈ 0/negative, so magnitude beats shape within the own-asset family, while no systematic co-movement metric adds signal (Figure 1 groups both axes). The increment is also not an artifact of the −100% delisting code. Mid-window delistings are ≤1% of asset-dates, and valuing them at last price or dropping them moves VaR's increment by ≤0.02 (Table 3). Table 2 gives the exact increments with BH significance.

VaR is worse than volatility standalone on equity (Table 1) yet has a positive increment there. As a standalone ranker it is a noisier proxy for volatility, but the partial correlation isolates its small orthogonal component. The increment is also larger where markets are fat-tailed and blow-up-prone (crypto increments 0.12–0.19, excess kurtosis 8–21, vs equity 0.03–0.08); a pooled per-date regression of the increment on tail-fatness across all 144 dates gives a positive slope, suggestive when clustered by bed. Tail magnitude carries incremental information precisely when the tail is active. Within crypto, splitting dates at their cross-sectional-median volatility, the increment rises from +0.13

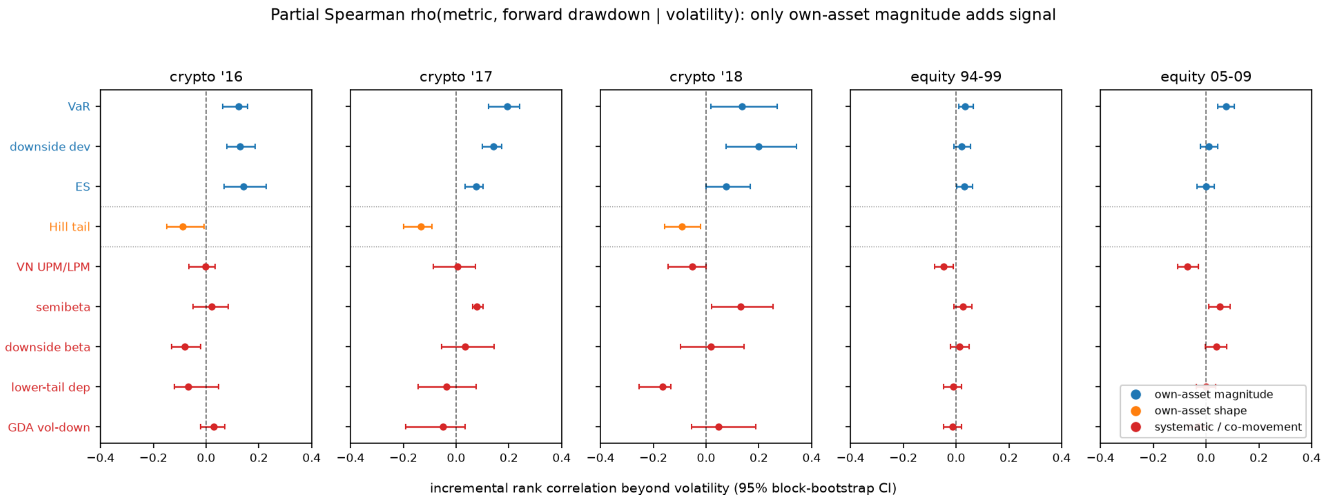

Figure 1: Incremental signal beyond volatility, by bed: partial Spearman ρ(metric, drawdown | volatility), grouped on both axes a skeptic cares about, namely what the metric measures (magnitude vs shape) and whose distribution it uses (own-asset vs systematic co-movement with the market). Only own-asset magnitude (blue) sits reliably right of zero. The lone own-asset shape metric, the Hill tail exponent (orange), is ≈ 0 or negative, so "magnitude beats shape" holds within the own-asset family, and no systematic co-movement metric (red) adds signal. Whiskers are 95% block-bootstrap CIs; dotted lines separate the three families.

Table 2: Partial Spearman ρ(metric, drawdown | volatility). Rows 1–3 tail magnitude, rows 4–9 tail shape. $*$ survives BH ($q = 0.05$, positive); † survives BH negative; n/e = not estimable per date. Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
VaR	+0.12$*$	**+0.19**$*$	+0.13$*$	**+0.03**$*$	**+0.08**$*$
downside dev.	+0.13$*$	+0.14$*$	**+0.20**$*$	+0.02	+0.01
ES [21]	**+0.14**$*$	+0.08$*$	+0.08	**+0.03**	+0.00
neg. semibeta [8]	+0.02	+0.08$*$	+0.13$*$	**+0.03**	+0.05$*$
UPM/LPM [23]	−0.00	+0.01	−0.05	−0.05†	−0.07†
downside beta [2]	−0.08†	+0.03	+0.02	+0.01	+0.04
Hill tail [17]	−0.09†	−0.13†	−0.09†	n/e	n/e
lower-tail dep. [9]	−0.07	−0.04	−0.17†	−0.01	+0.00
GDA [13]	+0.03	−0.05	+0.05	−0.01	−0.03

on low-stress dates to +0.18 on high-stress ones, while on equity it stays flat and small.

4.1 Is the target the reason?

Expected maximum drawdown scales with $\sigma\sqrt{T}$ [19], so drawdown is magnitude-dominated, and one might object that a magnitude metric is bound to win. The objection is correct, and it is the point. Drawdown is the loss that ruins investors, and the literature proposes shape metrics to forecast it. To confirm the shape metrics are not strawmen, we re-score every metric against genuinely shape-sensitive forward targets, namely forward return skew (sign-flipped so higher is more left-asymmetric), the downside fraction (forward downside deviation over forward

Table 3: VaR partial ρ(VaR, drawdown | vol) under three delisting codings. Baseline (−100%) and last-price coincide (mid-window delistings are rare); the increment is unchanged to within ±0.02.

Bed	baseline (−100%)	last-price	exclude
crypto-16	+0.12	+0.12	+0.14
crypto-17	+0.19	+0.19	+0.19
crypto-18	+0.14	+0.14	+0.14
equity-94	+0.035	+0.035	+0.035
equity-05	+0.076	+0.076	+0.076

volatility), and the tail-exceedance frequency (share of forward days below the trailing fifth percentile). The picture inverts (Table 4). Volatility's rank correlation on these targets is near zero or negative, while the shape metrics beat it on nearly every bed. Shape metrics forecast shape. Shape is simply not what drives drawdown. The daily adaptations are therefore faithful rather than weakened. The Hill exponent is a per-asset simplification of the Kelly–Jiang common tail factor, and the Viole–Nawrocki ratio is high when upside dominates downside, so its zero increment beyond volatility reflects a real null, not a wrong sign.

5 WHY VOLATILITY WINS, AND CAN ANYTHING BEAT IT?

5.1 Persistence

Volatility wins because it is persistent, the founding fact of ARCH/GARCH [7, 12]. Trailing volatility forecasts forward volatility (rank correlation 0.42–0.84), which in turn drives

Table 4: Winner by forward target. Volatility wins the magnitude target; shape metrics win every shape-sensitive target.

Forward target	vol rank corr (range)	shape beats vol
max drawdown (magnitude)	+0.36 to +0.62	none, any bed
−skew (shape)	−0.21 to −0.07	most, every bed
downside fraction (shape)	−0.25 to −0.05	most, every bed
tail-exceedance freq (shape)	−0.52 to −0.10	most, every bed

Table 5: Persistence chain (mean per-date Spearman; all CIs exclude 0).

Bed	trail→fwd vol	fwd vol→dd	(trail→dd)
crypto-16	0.68	0.77	0.62
crypto-17	0.42	0.62	0.36
crypto-18	0.42	0.53	0.38
equity 94	0.84	0.71	0.61
equity 05	0.80	0.70	0.56

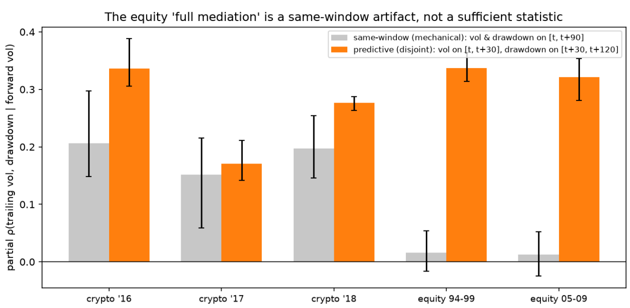

Figure 2: Mediation: partial ρ(trailing vol, drawdown | forward vol). The same-window (mechanical) design gives $\approx$ 0 on equity, falsely suggesting full mediation. The predictive (disjoint-window) design gives 0.32–0.34, so forward volatility does not fully screen off trailing volatility.

forward drawdown. Table 5 decomposes this chain by bed. Both links exceed the direct trailing-vol→drawdown rank correlation they compose. We resist calling forward volatility a *sufficient statistic*. A naive same-window mediation suggests it, but that test is contaminated (forward volatility and drawdown share a window). Measured on a disjoint, predictive window, trailing volatility retains a partial correlation of 0.32–0.34 on equity (Figure 2). Because that disjoint mediator is a noisy 30-day proxy, we read this as consistent with incomplete mediation rather than proof, so persistence is a leading but incomplete explanation, not a mechanism that reduces drawdown to one number.

Horizon dynamics. Persistence and forecastability rise and fall together. As we sweep the forecast horizon from 14 to 90 days, volatility persistence and the drawdown rank correlation move in step (Figure 3). On equity both rise (persistence 0.66 → 0.84, rank correlation 0.47 → 0.61) at a roughly constant gap, while on crypto both are flat. Wherever persistence is higher, so is the rank correlation, consistent with persistence driving most of the forecastable variation and the constant gap being the part it does not explain.

Table 6: Composite $z(\text{vol}) + z(\text{VaR})$ vs. volatility: per-date Spearman and difference (95% CI); significant only on crypto-bull.

Test bed	vol ρ	comp ρ	difference (95% CI)
crypto-16	0.62	0.62	−0.00 (n.s.)
crypto-17	0.36	0.40	+0.05 [0.02, 0.07]
crypto-18	0.38	0.39	+0.01 (n.s.)
equity 94	0.61	0.60	−0.008 [−0.011, −0.004]
equity 05	0.56	0.56	−0.00 (n.s.)
Locked panel: 0.376 → 0.388, Newey–West t = 0.67 (n.s.).			

5.2 Does the edge pay?

A rank correlation is not a profit-and-loss statement. The magnitude-not-shape split of §4, volatility plus a single tail-magnitude term, implies one natural composite, the equal-weight sum $z(\text{vol}) + z(\text{VaR})$ with no fitting, where $z(\cdot)$ is the cross-sectional z-score (each metric standardized across the assets on a given date). VaR was chosen a priori as the magnitude metric surviving correction on all five beds, and the weights are untuned, so the fully out-of-sample evidence is the locked panel. We pre-registered it. There it shows no significant out-of-sample gain over volatility (Newey–West t = 0.67, lag 3), beating it only on fat-tailed crypto-bull (Table 6). A cost-aware tercile backtest agrees (Figure 4). The high-risk tercile realizes 10–30 percentage points more forward drawdown than the low-risk tercile on every bed, net of a 20 bps turnover cost, and the composite adds nothing net of turnover except, again, on crypto-bull. Volatility is the workhorse. Magnitude adds a little, and only where tails are fat.

5.3 Can a machine find more?

If the residual is this small, a tireless automated search should fail out-of-sample and merely overfit validation. We test this on a strict train/validation/locked-test split, the locked test scored once after the metric is frozen. On the original locked panel, an LLM agent rewriting a scoring function and a 20,000-configuration sweep both inflate the best *validation* score with search intensity while the *locked-test* score plateaus at the volatility baseline (Table 7, Figure 5). Validation-selected winners beat volatility on the locked test only 43% of the time, despite a 0.94 validation–test rank correlation.

This is not a single-panel artifact. We re-ran a general nonlinear candidate generator (1,500 candidates per run) across a grid of holdout panels × 3 search seeds in two families: *out-of-time*, a fixed-length test window slid through the timeline with validation always chronologically earlier; and *random*, order-ignoring repartitions of the non-train dates. The validation-selected metric's locked-test edge over volatility (test gap) averages +0.020 out-of-time (range −0.028 to +0.046) and +0.035 random (+0.014 to +0.072), always small and often straddling zero, while the validation–test rank correlation stays 0.94–0.99 throughout. No

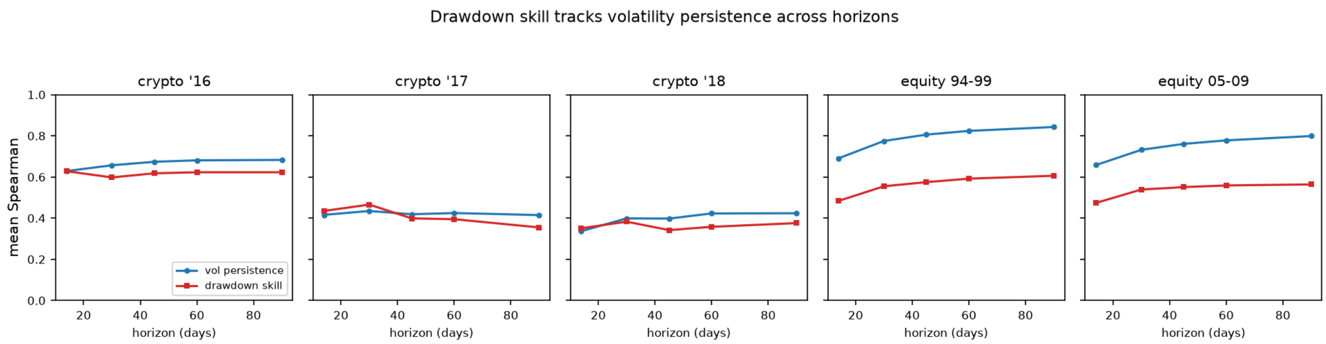

Figure 3: Volatility persistence (blue) and drawdown-forecast rank correlation (red) versus forecast horizon, by bed. The curves track.

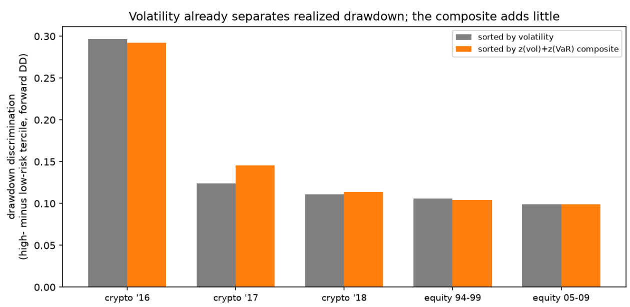

Figure 4: Realized drawdown discrimination (high minus low-risk tercile) by bed, sorting on volatility vs. the composite. Volatility already separates drawdown strongly; the composite is indistinguishable except marginally on crypto-bull.

Table 7: Best-of-N discovery on the original locked panel. Validation inflates with search intensity; the locked test does not (volatility baseline 0.376**). Validation and test are different splits, so compare trends, not levels.**

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373

search, LLM or brute-force, beats volatility out-of-sample by more than noise, and the non-LLM sweep reproduces the plateau, so the danger is the cheap search, not the language model.

The statistics are classical data snooping [5, 24]. What is new is that an autonomous agent removes the last brake on trial count, namely analyst effort, so a locked, survivorship-free holdout scored exactly once becomes mandatory, not merely advisable.

5.4 What is volatility for?

The clearest positive use is forecasting outright blow-ups. On the locked crypto test, volatility predicts which assets crater (forward return $\leq -80\%$) with ROC-AUC 0.68 (0.95 for the stricter $\leq -90\%$ cutoff) and a 2.5× top-decile lift, while a gradient-boosting model over all ten metrics, chosen on validation (AUC 0.73), overfits and loses on the locked test (AUC 0.59). The blow-up base rate is only $\approx$1.2%, so these AUCs are noisy and we read them as ordinal. Again the size, not the shape, of recent moves carries the signal. This is a ranking result, not a deployment recommendation (cost, liquidity, and reflexivity frictions are unmodeled).

6 HOW MUCH DO THE DISCIPLINES MATTER?

The three fair-field disciplines change what one would report, not the verdict. We ask which shape metric one would have credited without each.

Without survivorship-free data. We rebuild the crypto universe survivors-only, keeping at each formation date only coins that neither delisted nor ended below 5% of their all-time peak, about a third of the point-in-time top 100. On the 2018 crash bed this flips negative semibeta, a co-crash shape metric, from a head-to-head -0.067 against volatility (CI $[-0.131, -0.003]$, significantly worse) to $+0.055$ (CI $[-0.082, 0.178]$, a tie), closes downside beta from -0.23 to -0.01, and lifts VaR's partial from $+0.135$ to $+0.205$. Survivorship bias erases shape's penalty and inflates the magnitude edge, precisely in the crash regime where hiding the coins that died most reshapes the cross-section. It does not manufacture a winner. No shape metric beats volatility even survivors-only, and the rest stay far below it.

Without multiple-testing correction. Across the ~45 head-to-head cells the only metrics nominally beating volatility at $p < 0.05$ are downside deviation and VaR, both magnitude measures, and only on crypto-bull. No shape metric is even nominally positive, so Benjamini–Hochberg is not what holds shape back.

Without a locked holdout. The discovery experiment (§5) makes this concrete. A validation-selected metric's inflated validation score reads as a win, while its once-scored locked-test score sits at the volatility baseline. Even stripped of every guardrail, no shape metric beats volatility.

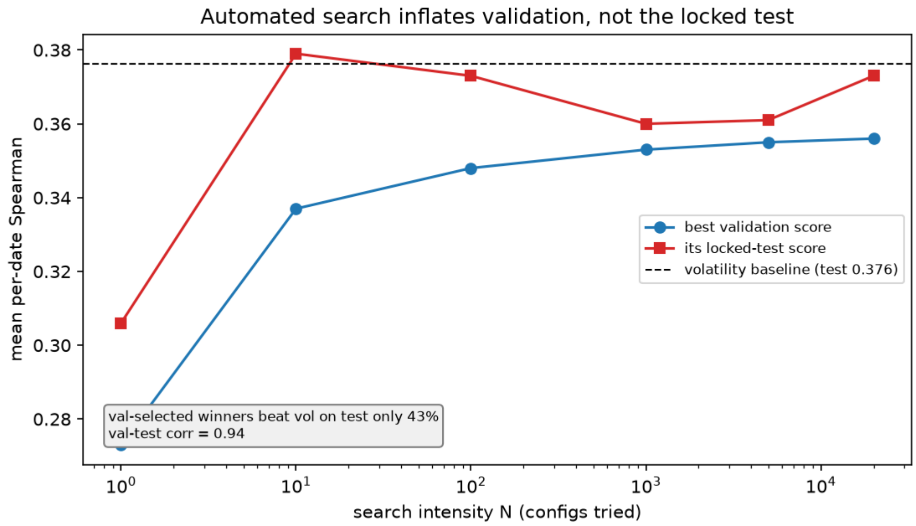

Figure 5: Automated discovery does not beat volatility out of sample. As search intensity grows, the best validation score climbs while its locked-test score stays flat at the volatility baseline. On this (2017 → 2018-crash) holdout, validation-selected winners beat volatility on the locked test only 43% of the time; across 12 panels × 3 seeds the win-rate is regime-dependent, but the locked-test edge over volatility never exceeds noise.

7 DISCUSSION AND CONCLUSION

The pieces form one coherent picture. Volatility sets a high bar because it is persistent (§5). The only thing left to add is the magnitude of tail losses, and only a little, mostly in fat-tailed markets (§4). So no standalone metric beats volatility (§3), a pre-registered composite yields no significant out-of-sample gain, automated search cannot find one, and the economics track the statistics (§5). The recurring failure of asymmetry and shape metrics (including the partial-moment ratio [23]) is not an implementation accident. Once symmetric volatility is accounted for, the asymmetric structure adds no incremental cross-sectional information for forecasting drawdown. The fair-field disciplines change what one would report, not the verdict (§6). Survivorship bias alone flips a co-crash shape metric from significantly worse than volatility to a tie and inflates the apparent magnitude edge, yet the verdict that no shape metric beats volatility is robust to dropping any single discipline. The disciplines guard the honesty of the numbers, not the direction of the conclusion.

We are deliberately careful about the strength of the persistence story. An earlier version of this study claimed forward volatility was a sufficient statistic, found that claim was a same-window measurement artifact, and retracted it here, just as an earlier "survivorship masks downside value" thesis was retracted when a hardened test refuted it. The paper reports only claims that survived stress-testing.

The practical takeaway is plain but firm. For cross-sectional drawdown forecasting, use volatility (optionally with a small tail-magnitude term in fat-tailed markets) and treat any downside metric "discovered" to beat it, by hand or by machine, as unproven until it clears a locked, survivorship-free holdout. On a level field, volatility is not glamorous, but it is very hard to beat.

Limitations. The performance measure is rank correlation, not a full tradable P&L (§5 is a first economic check, not a portfolio study); the crypto beds have few formation dates (7–9), so their significance is suggestive and the null "shape adds nothing" leans on the well-powered equity beds; and the repurposed pricing metrics are daily adaptations, so results speak to drawdown forecasting, not the originals' pricing claims.

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